

Computer Networks

Lecture 15: Data Link Layer

Content



- **Aloha Protocol**

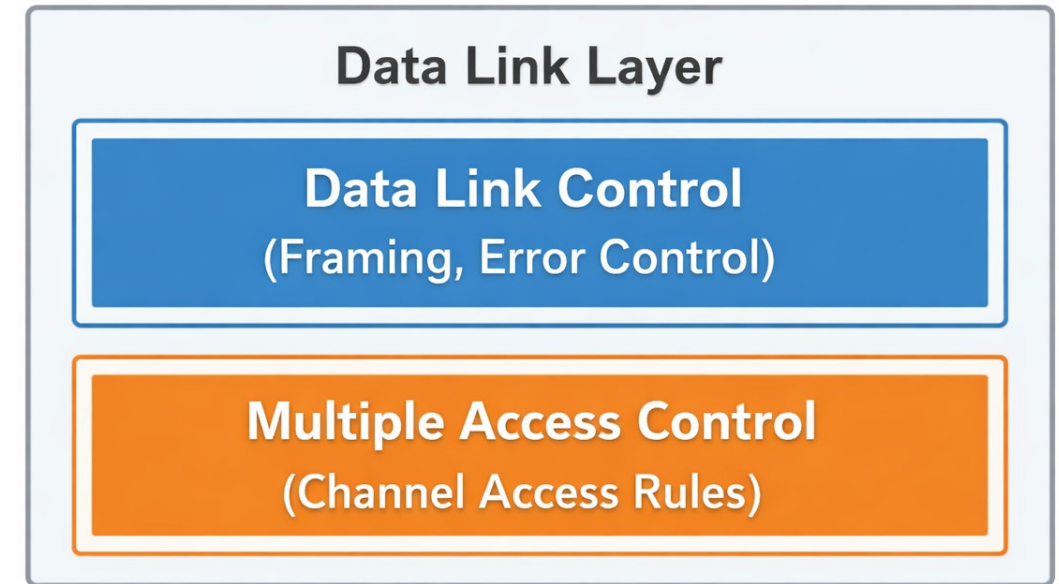
Introduction

We studied Data Link Control, which ensures reliable communication over a link. Those protocols assume a dedicated communication channel between sender and receiver

However, in many real-world systems (e.g., cellular networks), the channel is shared among multiple users.

Therefore,
the Data Link
Layer is
divided into
two
sublayers:

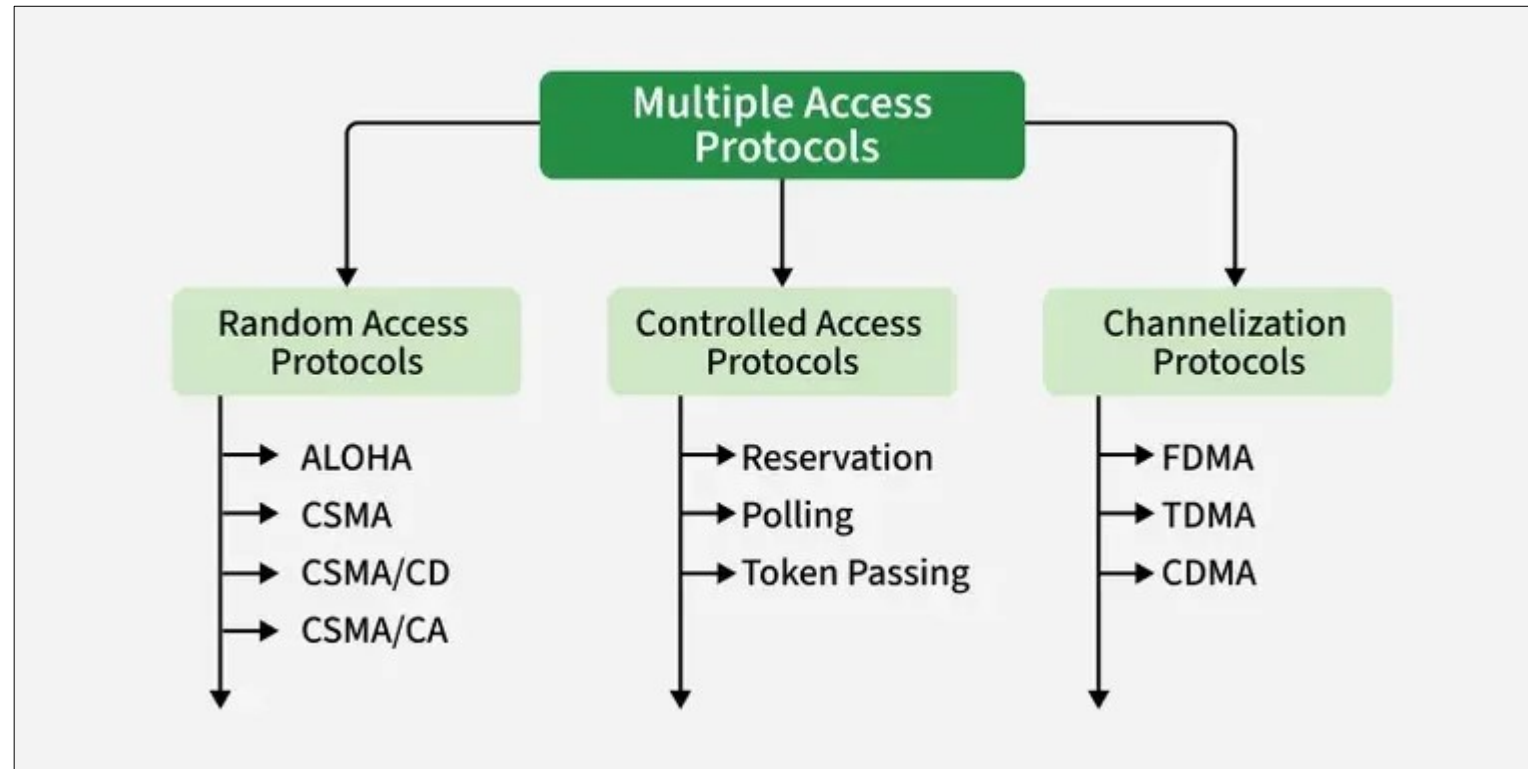
- **Upper Sublayer:** Data Link Control (reliability, framing, error control)
- **Lower Sublayer:** Multiple Access Control (MAC) – manages access to shared media



If the channel is dedicated, the MAC sublayer is not required.

Multiple Access Protocol

Many formal protocols have been devised to handle access to a shared link



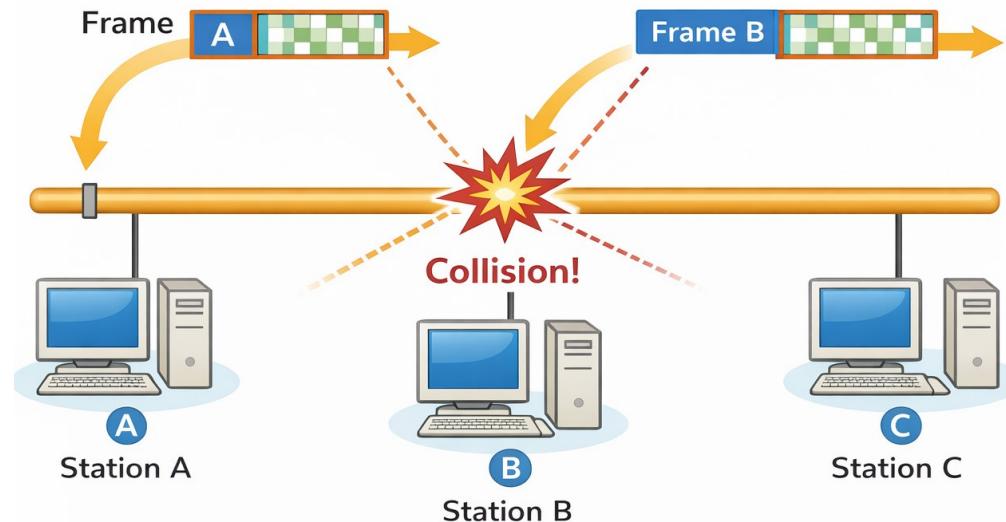
Random Access

In random access or contention methods,

No station is superior to another station and none is assigned the control over another.

No station permits, or does not permit, another station to send.

At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send. This decision depends on the state of the medium



Random Access

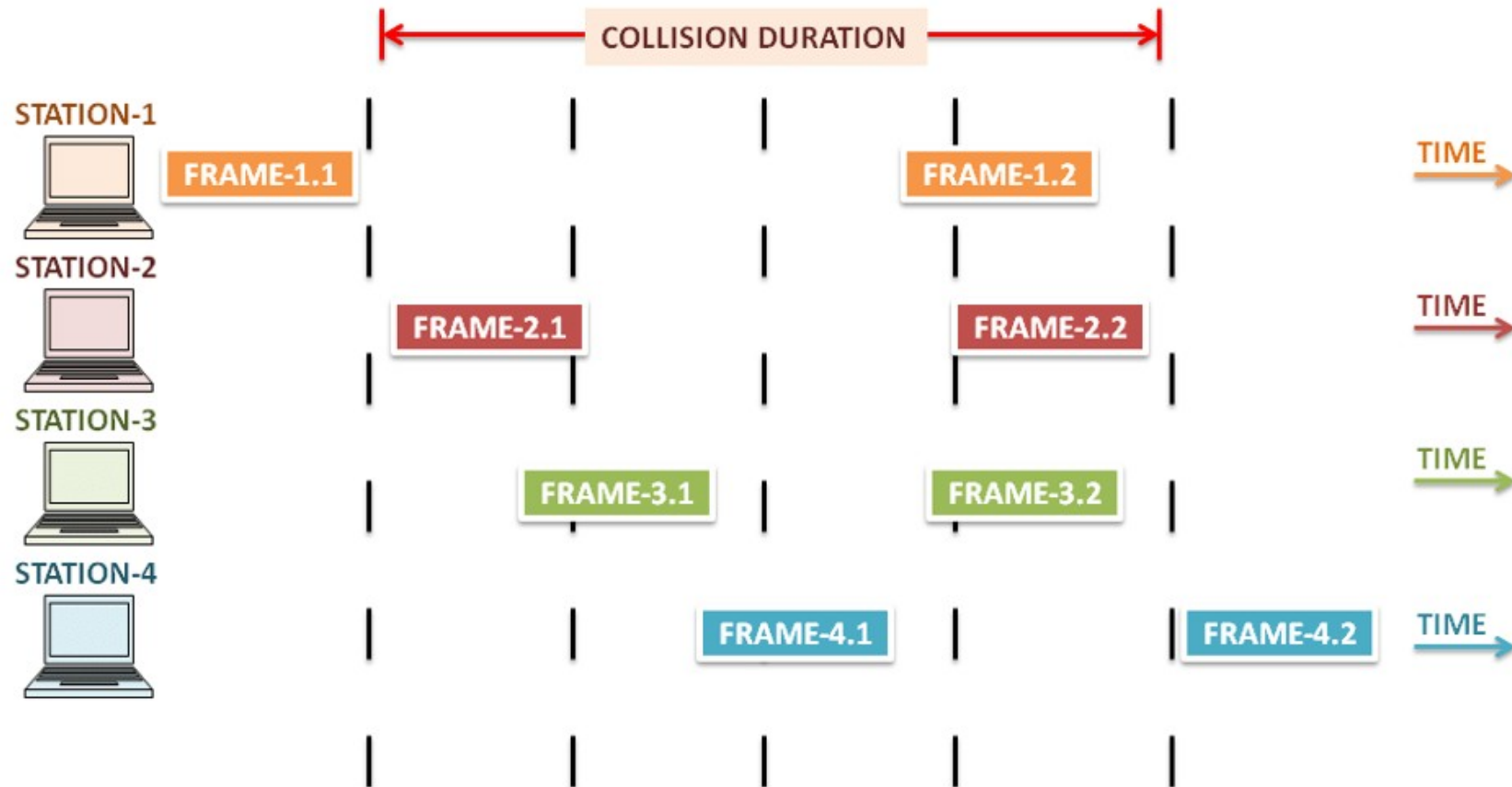
To avoid access conflict or to resolve it when it happens, each station follows a procedure that answers the following questions:

- When can the station access the medium?
- What can the station do if the medium is busy?
- How can the station determine the success or failure of the transmission?
- What can the station do if there is an access conflict?

Pure ALOHA

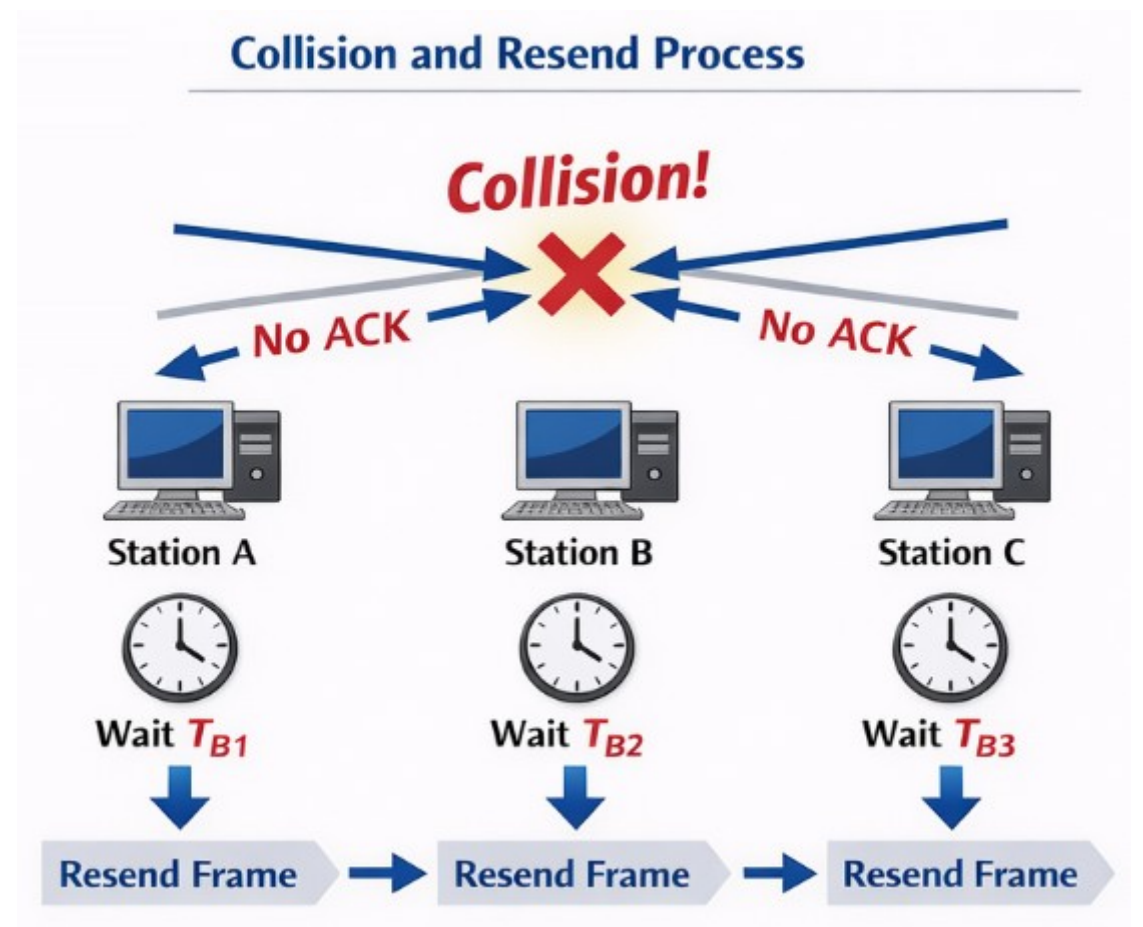
Frame Transmission

The idea is that each station sends a frame whenever it has a frame to send. However, since there is only one channel to share, there is the possibility of collision between frames from different stations.

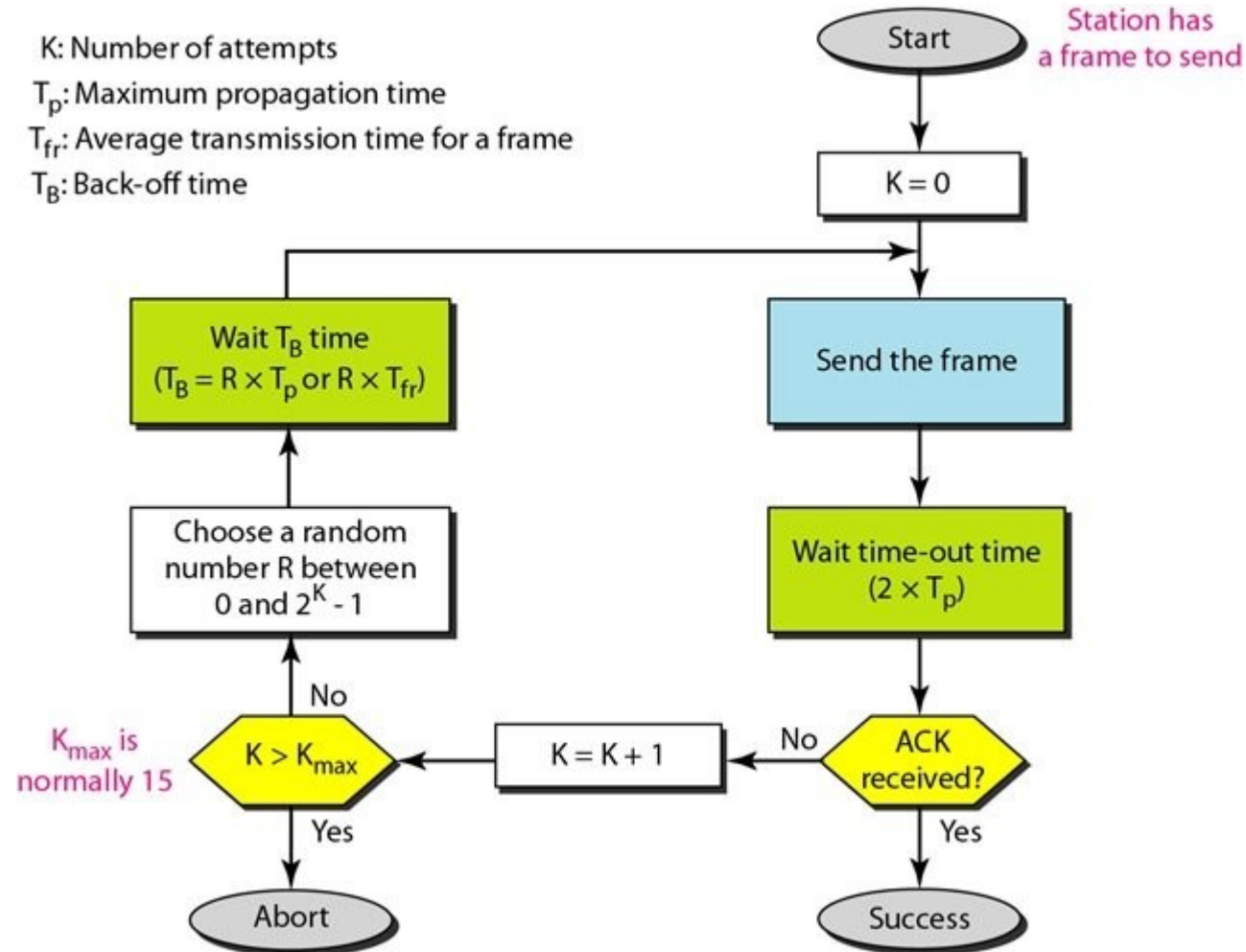


Pure ALOHA

- Frames lost during transmission must be retransmitted.
- After sending a frame, the sender waits for an **Acknowledgment (ACK)**.
- If no ACK is received before the **time-out**, the sender assumes collision or loss.
- The station waits for a **random Back-off Time (T_B)** before retransmitting to avoid repeated collisions.
- Random back-off reduces the probability of further collisions.
- After a maximum number of attempts ($K_{\max}$), the station stops retransmitting and tries again later.



Pure ALOHA



Question

The stations on a wireless ALOHA network are a maximum of 600 km apart. If we assume that signals propagate at 3×10^8 m/s, Find the value of T_B (back-off time) for the first three retransmissions.

- For $K = 1$, the range is $\{0, 1\}$. The station needs to generate a random number with a value of 0 or 1. This means that T_B is either 0 ms (0×2) or 2 ms (1×2), based on the outcome of the random variable.
- For $K = 2$, the range is $\{0, 1, 2, 3\}$. This means that T_B can be 0, 2, 4, or 6 ms, based on the outcome of the random variable.
- For $K = 3$, the range is to, $\{1, 2, 3, 4, 5, 6, 7\}$. This means that T_B can be 0, 2, 4, ..., 14 ms, based on the outcome of the random variable.

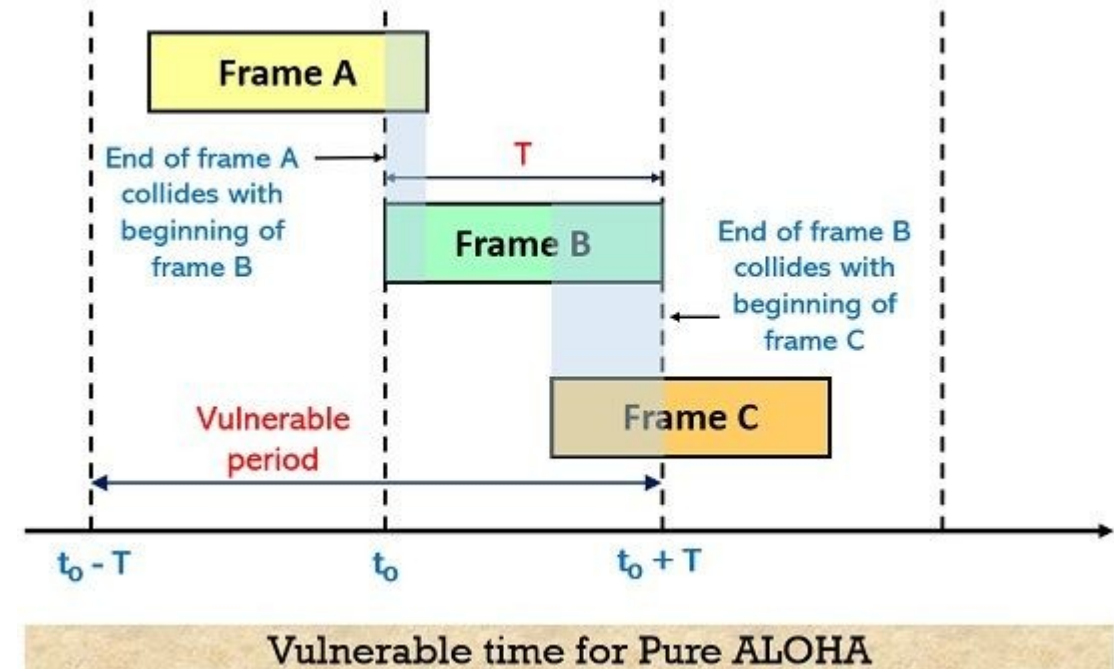
Pure ALOHA

Vulnerable time

Vulnerable Time is the period during which a transmitted frame is at risk of collision with other frames.

If a station starts transmitting at time t_0 :

- Any frame sent during the time interval $(t_0 - T)$ to $(t_0 + T)$ will collide with it.
- Therefore, the total vulnerable period is $2T$.



Throughput

In Pure ALOHA, a station transmits whenever it has data.
If two frames overlap in time



Assume that the Transmission attempts follow a Poisson process

$$P(k \text{ transmissions in time } t) = \frac{(\lambda t)^k e^{-\lambda t}}{K!}$$

- Consider that the average number of transmission attempts per frame time is G.
- Total vulnerable period is $2T$

Throughput

In order for any frame to be transmitted successfully no other station should be transmitting at that time so assume $K = 0$

$$P(\text{success}) = \frac{(2G)^0 e^{-2G}}{0!} = e^{-2G}$$

Thus, throughput is $G \times e^{-2G}$

Max, throughput is for $G=1/2$ which is approximately 0.184

Question

A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the requirement to make this frame collision-free?

- Average frame transmission time T_{fr} is 200 bits/200 kbps or 1-ms. The vulnerable time is $2 \times 1ms = 2ms$. This means no station should send later than 1-ms before this station starts transmission and no station should start sending during the one 1-ms period that this station is sending

In the last Question What is the throughput if the system (all stations together) produces

- a. 1000 frames per second
- b. 500 frames per second
- c. 250 frames per second